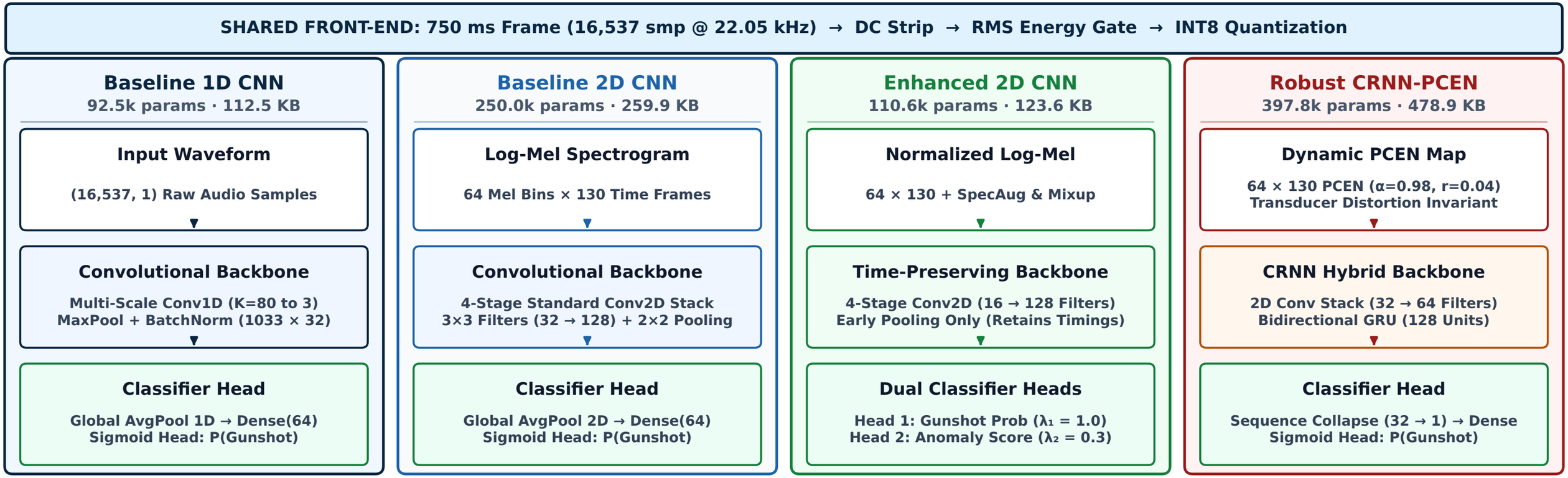
SHARED FRONT-END: 750 ms Frame (16,537 smp @ 22.05 kHz) → DC Strip → RMS Energy Gate → INT8 Quantization
Baseline 1D CNN
92.5k params · 112.5 KB
Input Waveform
(16,537, 1) Raw Audio Samples
Convolutional Backbone
Multi-Scale Conv1D (K=80 to 3)
MaxPool + BatchNorm (1033 × 32)
Classifier Head
Global AvgPool 1D → Dense(64)
Sigmoid Head: P(Gunshot)
Baseline 2D CNN
250.0k params · 259.9 KB
Log-Mel Spectrogram
64 Mel Bins × 130 Time Frames
Convolutional Backbone
4-Stage Standard Conv2D Stack
3×3 Filters (32 → 128) + 2×2 Pooling
Classifier Head
Global AvgPool 2D → Dense(64)
Sigmoid Head: P(Gunshot)
Enhanced 2D CNN
110.6k params · 123.6 KB
Normalized Log-Mel
64 × 130 + SpecAug & Mixup
Time-Preserving Backbone
4-Stage Conv2D (16 → 128 Filters)
Early Pooling Only (Retains Timings)
Dual Classifier Heads
Head 1: Gunshot Prob (λ₁ = 1.0)
Head 2: Anomaly Score (λ₂ = 0.3)
Robust CRNN-PCEN
397.8k params · 478.9 KB
Dynamic PCEN Map
64 × 130 PCEN (α=0.98, r=0.04)
Transducer Distortion Invariant
CRNN Hybrid Backbone
2D Conv Stack (32 → 64 Filters)
Bidirectional GRU (128 Units)
Classifier Head
Sequence Collapse (32 → 1) → Dense
Sigmoid Head: P(Gunshot)